

Shikha Nukala

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Education

University of Michigan, College of Engineering

Bachelor of Engineering in Computer Science

Ann Arbor, MI

May 2026

- Awards/Honors: Four terms Dean's List College of Engineering, University Honors
- Courses Completed: Foundations of Computer Science, Cryptography, Computer Security, Compilers, Computer Networks, Quantum Computing, Data Structures and Algorithms, Discrete Mathematics, Computer Organization

Relevant Projects and Experience

Forensics & DFIR:

- Conducted live and dead forensic analysis on a simulated compromised system using Autopsy and FTK Imager.
- Recovered deleted files, analyzed disk images, and bypassed security measures to uncover hidden artifacts and potential data exfiltration.
- Used cryptographic attacks, password crackers, network analysis, buffer overflow attacks, and web security vulnerabilities.
- Simulated SIEM tools to monitor web traffic, identify attack patterns, and detect SQLi, XSS, and buffer overflow attempts.
- Wrote a forensic investigation report in DFIR format, documenting attacks and mitigation strategies.

Network Defense & Threat Detection:

- Investigated simulated cyberattacks using Wireshark for packet analysis and abnormal activity detection.
- Used network traffic analysis to detect suspicious IPs, unauthorized access attempts, and potential exfiltration channels.
- Deployed Docker environments and Python sockets to simulate secure server-client communications and identify vulnerabilities.
- Configured IDS/IPS rules to alert on malicious traffic patterns such as DoS attempts.

Application Security & Reverse Engineering:

- Developed PowerShell and Bash scripts for log monitoring and detection of privilege escalation attempts.
- Analyzed binary exploits and memory forensics using Ghidra and GDB to understand overflow attacks and improve defenses.
- Used DEP with preset data for objects and removed extra memory allocation to prevent ROP attacks.
- Investigated malware samples and learned how to extract IoCs with reverse engineering.

Web Security:

- Hardened web applications against SQLi, XSS, and CSRF attacks by implementing input validation, secure authentication, and CSP policies.
- Designed security-enhanced authentication tokens to prevent session hijacking and replay attacks.

Cryptography Attacks & Exploits:

- Analyzed hashing flaws and cryptographic vulnerabilities and implemented secure encryption using the correct order of MAC, hashing, and RSA to uphold data integrity and confidentiality.
- Simulated attacks such as length extension to learn how to add secure encrypted padding.

Work Experience

Delta Dental of Michigan, Indiana, and Ohio

Application Development Intern

Okemos, MI

May 2024-Aug. 2024

- Converted SOAP APIs to REST and implemented secure authentication and authorization protocols to control access to provider data.
- Built an Angular application with input sanitization, HTTPS enforcement, and smoke tests to prevent injection attacks and ensure secure functionality.
- Collaborated with App Dev and product teams to perform API testing using Postman with secure token handling and error logging for threat detection.

Roosevelt Innovations

Knowledge Science Intern

Okemos, MI

May 2023-Nov. 2023

- Designed a secure Rules Engine using Drools and ANTLR, incorporating rule validation, execution sandboxing, and input constraints to prevent injection or logic manipulation.
- Modified a custom Angular IDE to guard against XSS through DOM sanitization, secure component rendering, and proper encoding of user-defined rules.
- Reduced third-party dependency in claims tools to minimize supply chain risks, while conducting internal audits of business logic for exploitation risks.

Leadership Experience

M-HEAL Team DRIP Technical Team Lead

Technical Team Lead

Ann Arbor, MI

Sept. 2024-present

- Leading website development for a hospital in Puerto Penasco, Mexico to raise awareness for diabetic amputees.
- Overseeing design and 3D printing of a prosthetic knee for diabetic amputees that is low-cost and highly durable.

Sahana Classical Dance and Music

Logistics Committee Lead

Ann Arbor, MI

Sept. 2022-Apr. 2024

- Arranged three hour dance competition shows for all Indian artistic groups at the University of Michigan.
- Booked and decorated Michigan theater, fundraised, coordinated performers, and sold 1000+ tickets.